Microsoft

Microsoft Learn
Spark possibility

Microsoft

AZ-040

Automate Administration with PowerShell

© Copyright Microsoft Corporation. All rights reserved.

Willkommen bei der TÜV Rheinland Akademie

Berlin Physik → Heidelberg

Thomas Jäkel

Lead Trainer Cloud Infrastructure

Microsoft Certified Trainer since 1999

NT 4.0

AD ~~Novell~~

github.com/www42/AZ-040

2011 Windows Azure

2013 Azure

Az

Az.*

TÜVRheinland®
Genau. Richtig.

M365

About this course

Overview

- This course teaches you how to use PowerShell to automate administration of Windows servers..

The Automating Administration with PowerShell course could be a great fit for you if you'd like to:

- Build scripts to accomplish advanced tasks such as automating repetitive tasks and generating reports.
- Automate administration across a broad range of Microsoft products, including Windows Server, Windows Client, Microsoft Azure, and Microsoft 365.

© Copyright Microsoft Corporation. All rights reserved.

Agenda

LP 1 Get started with Windows PowerShell

LP 2 Maintain system administration tasks in Windows PowerShell

LP 3 Work with the Windows PowerShell pipeline

LP 4 Work with PowerShell providers and PowerShell drives in Windows PowerShell

LP 5 Querying management information by using CIM and WMI

LP 6 Use variables, arrays, and hash tables in Windows PowerShell scripts

LP 7 Create and modify scripts by using Windows PowerShell

LP 8 Administer remote computers by using Windows PowerShell

LP 9 Manage cloud resources by using Windows PowerShell

LP 10 Manage Microsoft 365 services by using Windows PowerShell

LP 11 Create and manage background jobs and scheduled jobs in Windows PowerShell

5 7 ISE ... Get-Help

Labs

Item

Cert:

A: C: HKLM:

~~WMI~~

Modules

:5985

WinRM

WS-Man

Azure

EntraID 是 "Tenant"

© Copyright Microsoft Corporation. All rights reserved.

Let’s get to know each other

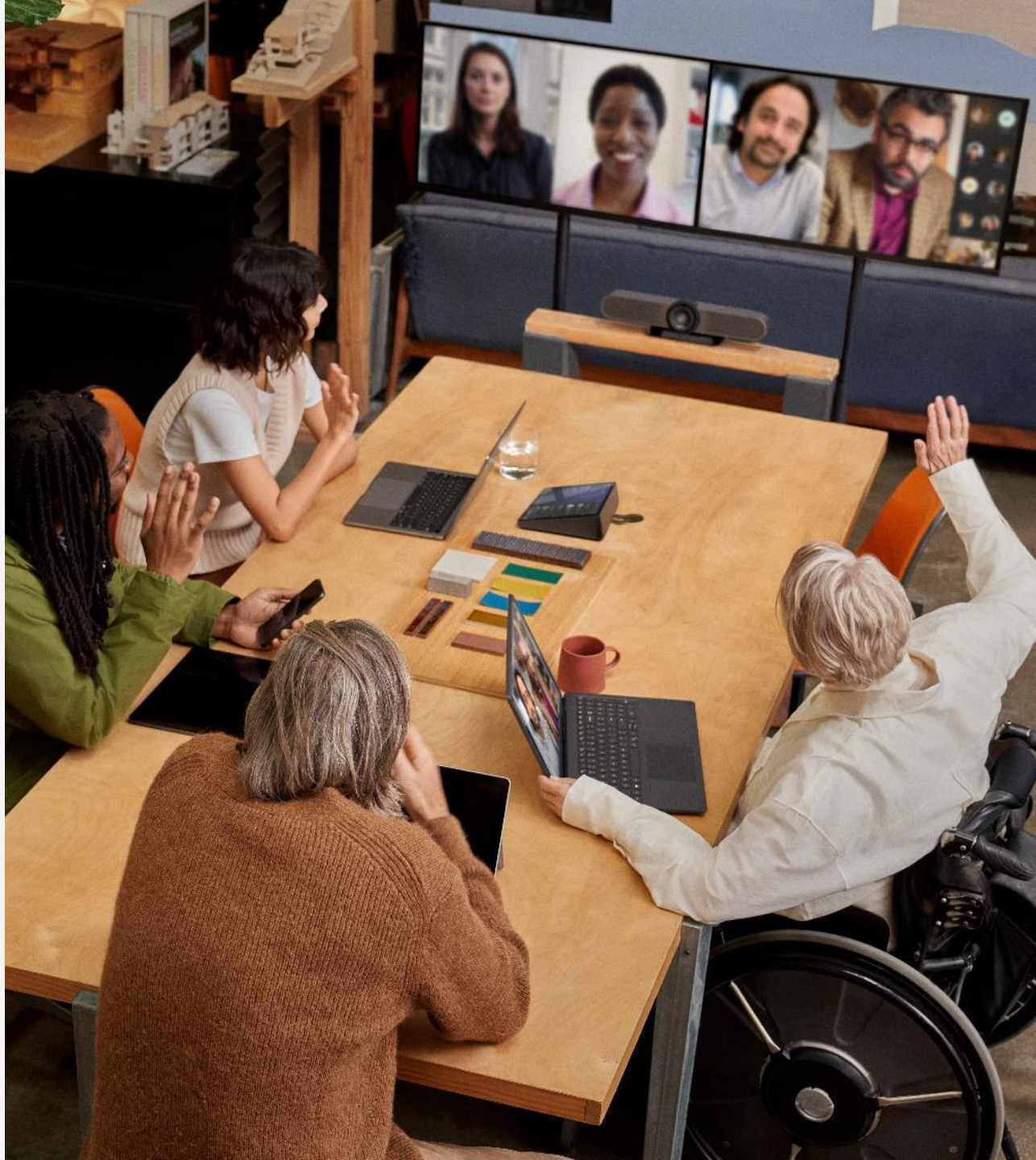
Your name

Company affiliation

Title/function

Your experience

Your expectations for the course



© Copyright Microsoft Corporation. All rights reserved.

Get the most out of your Microsoft Learn profile

Verify, track, and share your training and certification progress and accomplishments—*all on one platform*

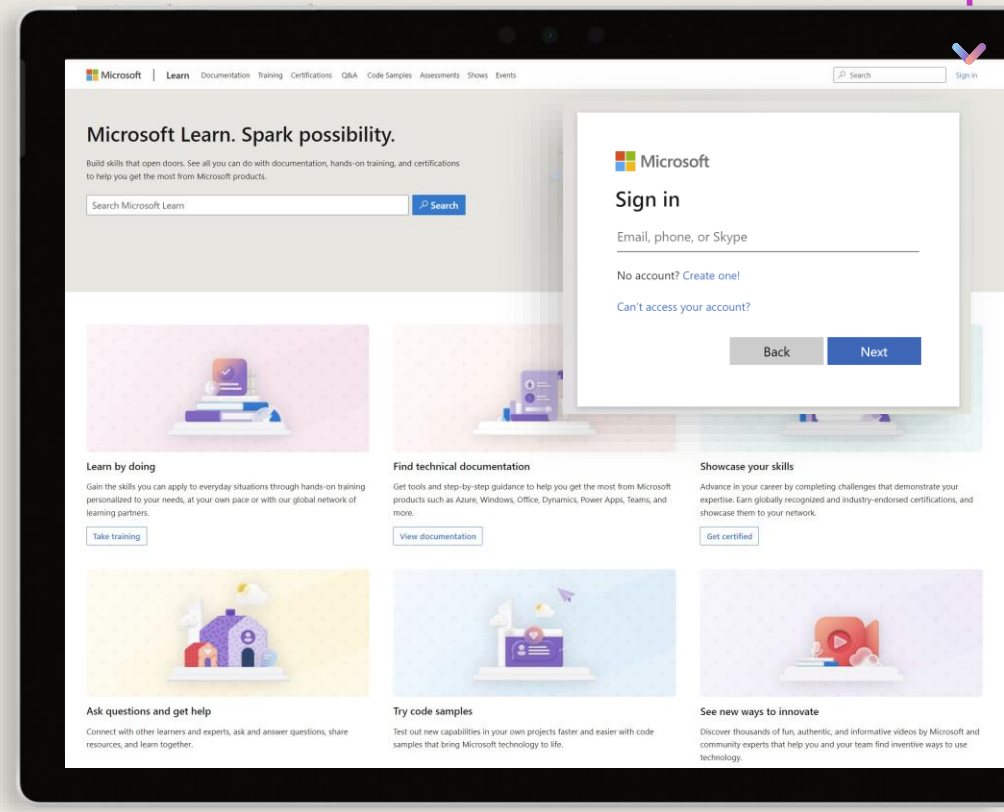
- Claim your achievement code for this course and share you have completed it.
- Access your course material and track progress on your learning activities.
- Share and verify your Microsoft Certifications via email, on social networking platforms, and on your résumé.
- Download and print transcripts and certificates.
- Manage your upcoming activities and certification exam appointments.

www.aka.ms/MyMicrosoftLearnProfile

© Copyright Microsoft Corporation. All rights reserved.

Create your Microsoft Learn profile at learn.microsoft.com

- Select *Sign in* at the top, right corner of any Microsoft Learn page.
- Follow the Microsoft account authentication process.
- If the account that you have chosen to sign-in with doesn't already have a Microsoft Learn profile, you'll be guided to create one.



Let’s have a great time together

We all contribute to a great class

- Be on time (class hours)
- Be present (avoid mail, calls, side meetings)

$9^{00} - 17^{00}$

$12^{30} - 13^{15}$

15 min

What you should know about our facilities

- Internet access
- Parking
- Restrooms
- Emergency

© Copyright Microsoft Corporation. All rights reserved.

go deploy - Lab Umgebung

go deploy AZ-040T00 (CS) | Lab 01

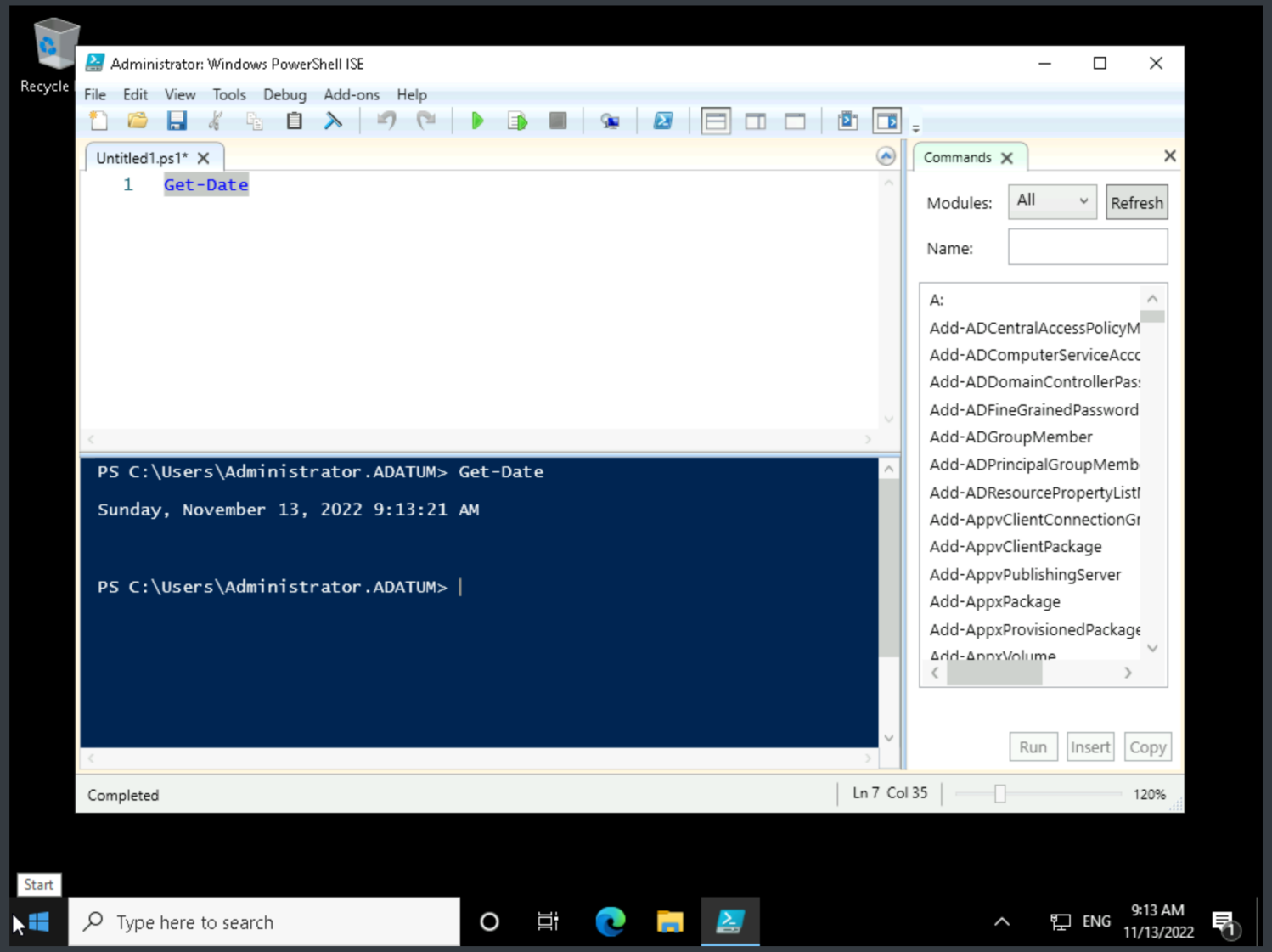
Home | Lab Guide | Microsoft Learn

Lab Guide

Task 1: Start the console application as Administrator, and pin the Windows PowerShell icon to the taskbar

- [x] 1. On LON-CL1, send the CTRL+ALT+DEL command and then log in as ADATUM\Administrator with the password Pa55w.rd.
- [x] 2. Select **Start**.
- [ ] 3. Enter **powershell** to display the Windows PowerShell icon. Make sure that the icon name displays **Windows PowerShell** and not **Windows PowerShell (x86)**.
- [ ] 4. Right-click **Windows PowerShell** or activate its context menu, and then select **Run as administrator**.
- [ ] 5. Make sure that the window title bar reads **Administrator** and doesn't include the text **(x86)**. This indicates that it is the 64-bit console application and that an administrator is running it.
- [ ] 6. On the taskbar, right-click the **Windows PowerShell** icon or

00:16:21



Skillable - Lab Umgebung

AZ-040T00 Automating Administration with Windows PowerShell - Personal - Microsoft Edge

https://labclient.labondemand.com/LabClient/8a3545a1-29f9-4f0e-9d59-bd8db8b88d5b

AZ-040T00A-LON-CL1

Recycle Bin

Administrator: Windows PowerShell

```
PS C:\> $PSVersionTable

Name                           Value
----                           -----
PSVersion                      5.1.19041.1237
PSEdition                      Desktop
PSCompatibleVersions           {1.0, 2.0, 3.0, 4.0...}
BuildVersion                   10.0.19041.1237
CLRVersion                     4.0.30319.42000
WSManStackVersion              3.0
PSRemotingProtocolVersion      2.3
SerializationVersion           1.1.0.1


PS C:\>
```

Type here to search

DE 12:30 PM 1/25/2026

AZ-040T00 Automating Administration with Windows PowerShell

Exit Lab

Instructions Resources

Exercise 1: Configuring the Windows PowerShell console application

Task 1: Start the console application as Administrator, and pin the Windows PowerShell icon to the taskbar

1. On AZ-040T00A-LON-CL1, select **Start**.
2. Enter powershell to display the Windows PowerShell icon. Make sure that the icon name displays **Windows PowerShell** and not **Windows PowerShell (x86)**.
3. Right-click **Windows PowerShell** or activate its context menu, and then select **Run as administrator**.
4. Make sure that the window title bar reads **Administrator** and doesn't include the text **(x86)**. This indicates that it is the 64-bit console application and that an administrator is running it.
5. On the taskbar, right-click the **Windows PowerShell** icon or activate its context menu, and then select **Pin to taskbar**. The **Windows PowerShell console** should now be open, run by **Administrator**, and available on the taskbar for future use.

Task 2: Configure the Windows PowerShell console application

1. To configure Windows PowerShell to use the **Consolas** font:

   a. Select the control box in the upper-left corner of the **Windows PowerShell console** window.

Previous Next

29 Hr 54 Min Remaining